University Physics: Mechanics QUIZ

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] A cart starts at v■=3 m/s and accelerates at 3 m/s^2 for 2 s. Find final velocity and displacement; include units.

2. [8 pts] A 7 kg block is pulled horizontally by 27 N on a surface with μ_k=0.2. Draw the free-body diagram and compute acceleration (g=9.8 m/s^2).

3. [7 pts] A 2 kg object drops from height 8 m with negligible drag. Use energy conservation to find its speed just before impact.

4. [5 pts] State momentum conservation for a one-dimensional collision and identify the condition under which it applies.

5. [6 pts] A force-versus-time graph has a rectangular pulse of height 6 N lasting 0.5 s. Compute impulse and describe its effect on momentum.